Certified genuine multipartite entanglement of a 24-qubit GHZ state on ibm_fez, found by an autonomous AI scientist

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ABSTRACT

This paper reports an autonomous search for large Greenberger–Horne–Zeilinger (GHZ) states conducted on the 156-qubit IBM processor ibm_fez. A language-model agent (gemini-2.5-flash) designed 8 hardware experiments in a single planning step, followed by a second round of 4 experiments after processing initial results. Adaptive scheduling was not feasible due to the public queue. The second round included a pre-registered confirmation of a prior result. Prior to submission, each experiment was simulated using a digital twin, which is a noise model informed by live calibration data. The total QPU time consumed was 322 s, within a self-imposed budget of 480 s.

Fidelity to the ideal GHZ state was estimated using population and parity-oscillation measurements, with bootstrap confidence intervals. Genuine multipartite entanglement was claimed only when the lower confidence bound exceeded 1/2. A GHZ state of N = 24 qubits was certified on hardware with a fidelity F = 0.723 (CI 0.700–0.746). This N = 24 state was certified in 2 separate hardware experiments utilizing different tree layouts on overlapping qubits (20 of 24 shared). An N = 40 result (F = 0.654, CI 0.520–0.788) was not claimed, as its pre-registered confirmation on the same qubits yielded F = 0.414 (CI 0.326–0.486) with 2000 shots. For N = 24, a shallow tree circuit (two-qubit depth 8) achieved F = 0.723, whereas a linear CNOT chain (depth 23) on the same device yielded F = 0.107; only the tree layout was certified.

The digital twin, whose predictions were fixed before experiment submission, consistently overestimated hardware fidelity across all 12 experiments (mean +0.238, range +0.003 to +0.597), indicating unmodeled noise. This work discusses the statistical robustness of the reported results, the influence of error mitigation techniques, and the limitations inherent in an agent selecting its own optimal outcome. This paper was generated automatically from the run log.

1 INTRODUCTION

Greenberger–Horne–Zeilinger (GHZ) states [1, 2], $|\mathrm{GHZ_N}\rangle = (|0\rangle^{\otimes N} + |1\rangle^{\otimes N})/\sqrt{2}$, are the canonical maximally entangled states of N qubits. Because their single coherence term $|0\ldots0\rangle\langle1\ldots1|$ dephases N times faster than a single qubit, the largest GHZ state a device can prepare is a compact, demanding benchmark of its two-qubit gates, readout and coherence all at once. GHZ states of increasing size have marked progress across platforms, from trapped ions [5] to Rydberg atoms [6] and superconducting circuits [7, 8], and today's processors with more than a hundred qubits [9] make much larger states conceivable.

Preparing a state is not the same as proving it is entangled. A standard witness [3, 4] uses the fidelity $F = \langle \mathrm{GHZ_N}|\rho|\mathrm{GHZ_N}\rangle$: any state with $F > 1/2$ is *genuinely multipartite entangled* (GME), i.e. it cannot be written as a mixture of states that are separable across any bipartition. We therefore treat certification as a statistical claim: the lower end of a confidence interval on F must exceed 1/2.

Scaling the certified N on a 156-qubit heavy-hex processor [10] is a search problem over layouts, circuit shapes and mitigation settings under a tight hardware budget. In this work the search was run end-to-end by an autonomous agent built on a large language model (Gemini), which proposed every experiment, predicted each hardware result with a digital twin before it ran, and wrote this report. The human operator set the goal and budget and ran the submit and collect commands; which experiments to run was the agent's choice.

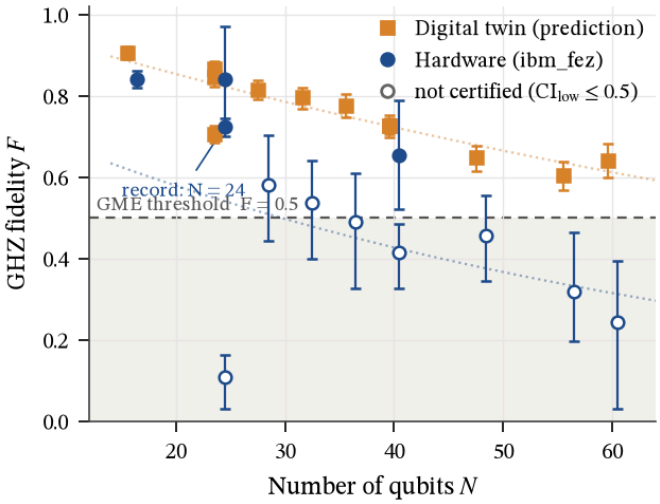


Figure 1. GHZ fidelity versus size for every successful experiment. Error bars: bootstrap confidence intervals. Filled markers are certified (lower bound above 1/2); open markers are not. Dotted lines: exponential fits $F_0 e^{-N/N_e}$. Shaded: region below the GME threshold.

2 METHODS

2.1 Hardware and calibration

Experiments targeted **ibm_fez**, a 156-qubit IBM Heron processor with 176 tunable couplers arranged in a heavy-hex lattice and native CZ entangling gates. At the start of the run the agent recorded a live calibration query (timestamp 2026-11-08T03:07:49+00:00). Median values across the chip were $T_1 = 148$ μs, $T_2 = 107$ μs, readout error 1.07%, single-qubit ($\sqrt{X}$) error 2.9×10^{-4} and two-qubit error 0.28%. The snapshot (Fig. 3) drove layout selection and the noise model of the simulator: a Pauli-frame simulation (depolarizing gate errors, per-layer T_1/T_2 idle decoherence from real gate durations, and readout error) that keeps the per-qubit and per-coupler noise of the physical qubits actually used. Hardware runs, where present, need no such approximation.

2.2 Layouts and GHZ circuits

A GHZ state on N physical qubits is prepared by a Hadamard on a root qubit followed by CNOTs (compiled to CZ) fanning out along a spanning tree of the coupling graph, so the entangling depth scales with the tree height rather than with N. The agent chose among layout strategies: *best_chain* selects a low-error one-dimensional path through the coupling graph; *bfs_tree* takes a breadth-first tree around the root, minimising tree height and hence circuit depth without regard to error rates; *weighted_tree* grows a spanning tree outward from a root qubit, at each step adding the neighbouring qubit with the lowest combined cost of CZ error, readout error and dephasing (from the calibration snapshot), with a penalty on tree depth, which steers the state away from poorly calibrated couplers. It could also exclude specific qubits and select a linear (chain) circuit instead of a tree. For the record experiment (e002, N = 24) the tree has 23 two-qubit gates arranged in 8 layers of two-qubit-gate depth (total transpiled depth 32).

2.3 Fidelity estimation

Following Refs. [6, 8], the fidelity splits into a population and a coherence term,

$$F = \tfrac{1}{2}(P + C), \quad P = p(0\ldots0) + p(1\ldots1), \tag{1}$$

where P is read off a computational-basis measurement. The coherence $C = 2|\rho_{0\ldots0,1\ldots1}|$ is obtained from parity oscillations: after preparing the state, every qubit receives a phase rotation $R_z(\varphi)$ and is measured in the X basis, and the parity

$$\langle \mathcal{P}(\varphi)\rangle = \Sigma_x (-1)^{|x|}\, p_\varphi(x) = C\cos(N\varphi + \delta) \tag{2}$$

is sampled at the $2N + 2$ phases $\varphi_j = j\pi/(N + 1)$, $j = 0,\ldots,2N{+}1$. Only a GHZ-type coherence oscillates at frequency N, so C is its Fourier amplitude,

$$C = \frac{1}{N{+}1}\,\left|\,\Sigma_j\, e^{iN\varphi_j}\,\langle \mathcal{P}(\varphi_j)\rangle\,\right|. \tag{3}$$

Each experiment therefore runs $2N + 3$ circuits (1000 or 2000 shots each), plus two readout-calibration circuits when mitigation is on. A 95% confidence interval on F comes from a percentile bootstrap that resamples the shots of every circuit independently [13], and an experiment is *certified* when $F_{\text{low}} > 1/2$, which by the witness of Refs. [3, 4] implies genuine multipartite entanglement of the prepared state. Error suppression and mitigation: readout-error mitigation in 100% of experiments, inverting per-qubit (tensored) confusion matrices estimated from all-$|0\rangle$ and all-$|1\rangle$ calibration circuits; because both the GHZ populations and the parity are product-form functionals of the bitstring, this inversion is applied exactly, shot by shot; on hardware experiments, XpXm dynamical decoupling on idle qubits [11] (100%) and measurement twirling [12] (100%) via the Qiskit Runtime sampler; these are not modelled in simulation. Circuits were built and transpiled with Qiskit [14], keeping the planned physical layout fixed.

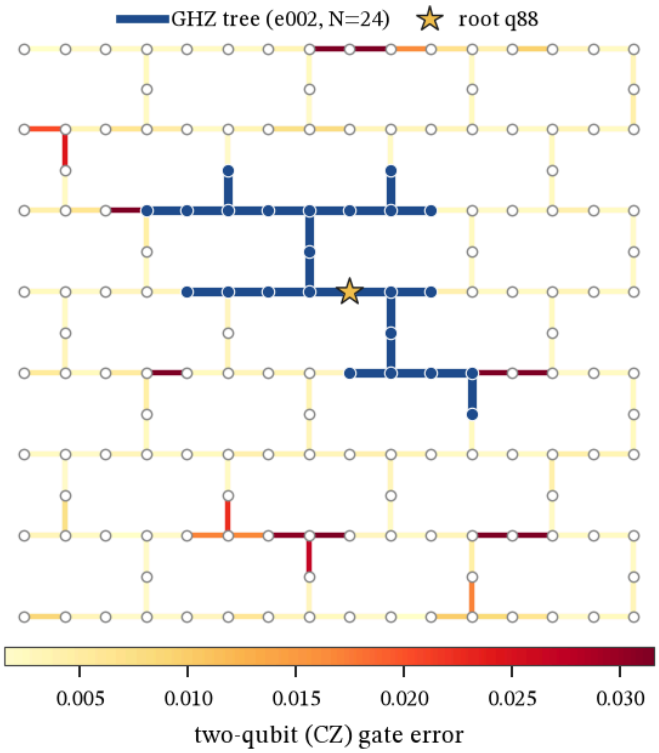


Figure 3. Calibration snapshot of ibm_fez: couplers coloured by CZ error, with the GHZ tree of e002 highlighted (root ★).

3 THE AUTONOMOUS AGENT

In its interactive mode the agent runs a closed loop over simulator and hardware experiments. On the public queue of ibm_fez, however, a single job can wait hours, so a loop that waits for each result could finish only one or two hardware experiments in a night. For this run the agent therefore designed each round of hardware experiments in a single planning step. Its context was the live calibration summary, its own earlier simulator results and a cost table for each ex-

periment size. Every proposed experiment was then checked by a validator, which enforces a safe small-N anchor, a matched tree-versus-chain comparison and the QPU budget. Each experiment was also run through a digital twin (the noise simulator of §2.1 driven by the *live* calibration) to record a prediction *before* the hardware result existed. All jobs were then submitted at once and collected as they completed.

The experiments were: round 1, e001: N = 16 weighted_tree; e002: N = 24 weighted_tree; e003: N = 24 best_chain; e004: N = 32 weighted_tree; e005: N = 40 weighted_tree; e006: N = 48 weighted_tree; e007: N = 56 weighted_tree; e008: N = 60 weighted_tree. Round 2, designed after reading every round-1 result, with the validator requiring a pre-registered re-run of the largest certified result: e009: N = 40 weighted_tree (re-run of e005); e010: N = 36 weighted_tree; e011: N = 24 bfs_tree; e012: N = 28 weighted_tree. Every model call was charged against a hard cap before it was made; the run used 322 of its 480 self-imposed QPU seconds (the open plan allows 600). The agent's round-1 rationale, verbatim:

> *"This batch is designed to meet all requirements: a robust anchor, a direct comparison between 'weighted_tree' and 'best_chain', and an exploration of the hardware frontier for 'weighted_tree'. The N=16 anchor ensures a certified result even if larger experiments fail. The N=24 matched pair will directly assess if the error-aware 'weighted_tree' layout outperforms the textbook 'best_chain' on real hardware. The remaining experiments progressively increase N for 'weighted_tree' up to N=60, aiming to identify the maximum number of qubits for certified entanglement on ibm_fez, anticipating a lower frontier than simulator results due to real-world noise. All experiments use 1000 shots to maximize confidence in fidelity estimation within the budget."*
>
> — agent, batch plan

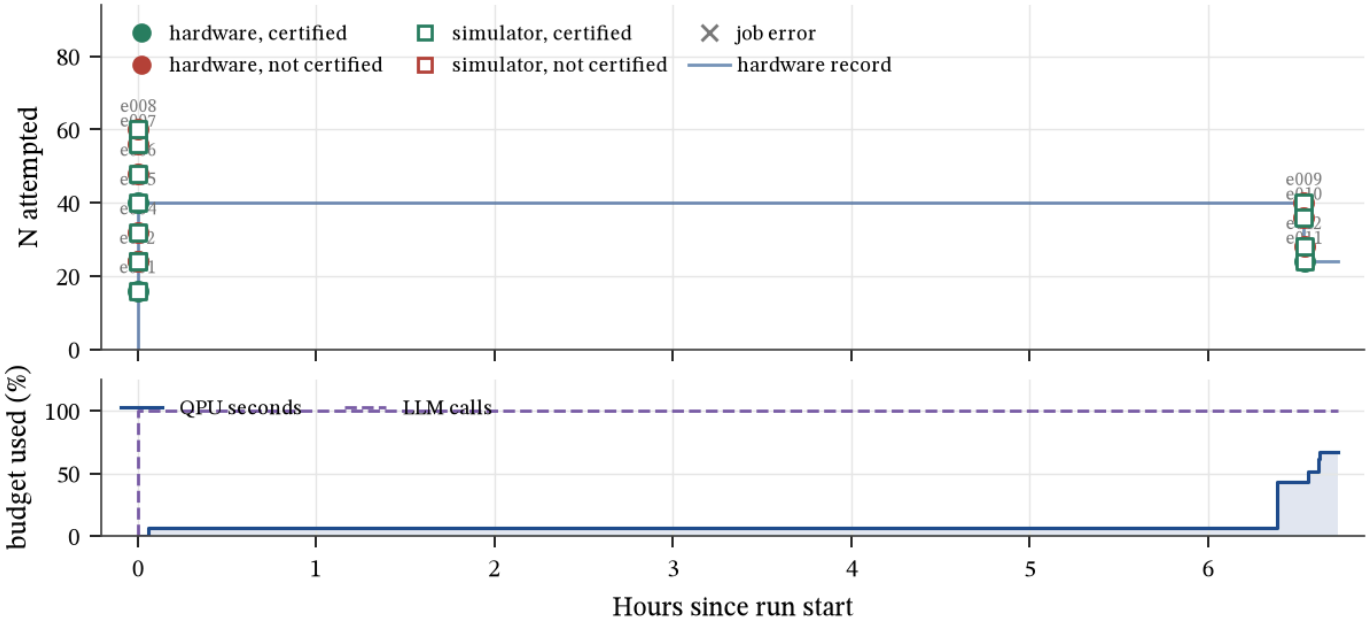


Figure 4. Timeline of the run. Top: qubit number of each experiment against wall-clock time, coloured by outcome; the line tracks the certified hardware record. Bottom: cumulative fraction of the QPU-time and LLM-call budgets consumed.

4 RESULTS

The largest state certified on hardware is the **N = 24** GHZ state of experiment e002 (Table 1), with $F = 0.723$ and confidence interval [0.700, 0.746]: the lower bound clears the GME threshold by 0.200. Its populations and coherence are $P = 0.743$ and $C = 0.703$; as is typical, the coherence term limits the fidelity. The state used a weighted_tree layout rooted at q88 (Fig. 3) whose couplers have a mean CZ error of 0.30% and mean readout error 1.06%, against chip medians of 0.28% and 1.07%.

Readout mitigation and noise. Inverting the readout model amplifies shot noise as N grows: across the run, individual mitigated parities reach $|\langle P \rangle| = 4.22$, outside the physical range ±1. The second N = 24 certification, e011 ($F = 0.842$ [0.720, 0.972]), has mitigated parities up to 1.75 and an interval 6× wider than e002's, so e002 is quoted as the record. The certified values are properties of the mitigated estimator; an unmitigated check was not performed.

A withdrawn record. In the first round, e005 certified an N = 40 state with $F = 0.654$ [0.520, 0.788], a lower bound only 0.020 above the threshold, while the smaller N = 32, 36, 28 tree states did not certify. Its validator-enforced confirmation, e009, re-ran the identical specification on the same physical qubits with 2000 shots and measured $F = 0.414$ [0.326, 0.486]: not certified. N = 40 is therefore not claimed; the first result is best read as an upward fluctuation selected

from several noisy attempts, possibly compounded by calibration drift between the two runs. This is the selection effect discussed in Section 5, observed directly.

Figure 2 shows the parity oscillation of e002. For a GHZ-type coherence, folding the 50 phase settings by $N\varphi$ collapses them onto a single cosine, the signature of an N-body coherence. An independent least-squares fit of Eq. (2) gives amplitude 0.703 (residual RMS 0.074), consistent with the estimator value $C = 0.703$.

Fidelity decays roughly exponentially with size (Fig. 1). Fitting $F(N) = F_0\, e^{-N/N_e}$ to the central values gives hardware: $N_e \approx 66$, crossing $F = 1/2$ near $N \approx 30$; simulation: $N_e \approx 120$, crossing $F = 1/2$ near $N \approx 84$. These fits are descriptive only; they ignore the different layouts used at each size.

Tree circuit versus linear chain. At $N = 24$ the error-weighted tree (e002, two-qubit depth 8) reached $F = 0.723$ [0.700, 0.746], against 0.107 [0.031, 0.163] for the linear CNOT chain (e003, depth 23), a difference of +0.616. Only the tree certified GME. The twin had predicted 0.866 and 0.705 respectively, so the sign of the difference was correctly anticipated before the hardware ran. A breadth-first tree that ignores error rates (e011) reached $F = 0.842$ at the same size, so the advantage over the chain comes from circuit depth rather than from error-weighting the layout. (Fig. 5)

Prediction versus hardware. Across 12 hardware experiments the digital twin overestimated fidelity by 0.238 on average (range +0.003 to +0.597) and predicted the certification outcome correctly in 4 of 12 cases. Because every prediction was recorded before submission, this is an honest out-of-sample test of the noise model.

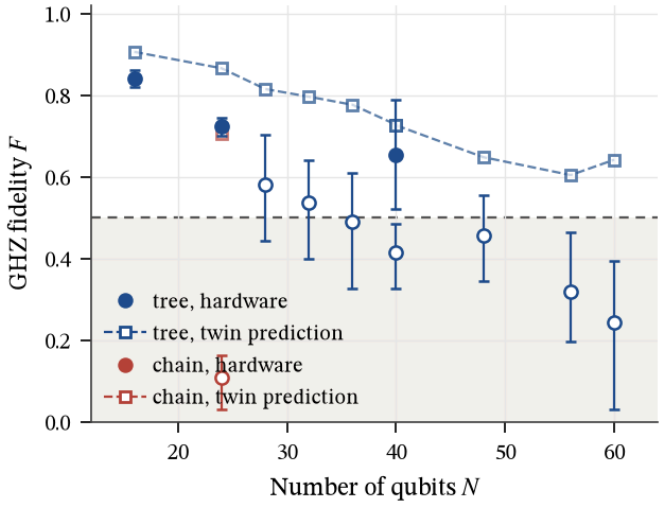


Figure 5. Hardware fidelity (circles, bootstrap CI; filled = certified) against the digital-twin prediction recorded before submission (squares, dashed), for the error-aware tree and the textbook linear chain.

Figure 4 places every experiment on the run's timeline together with budget consumption.

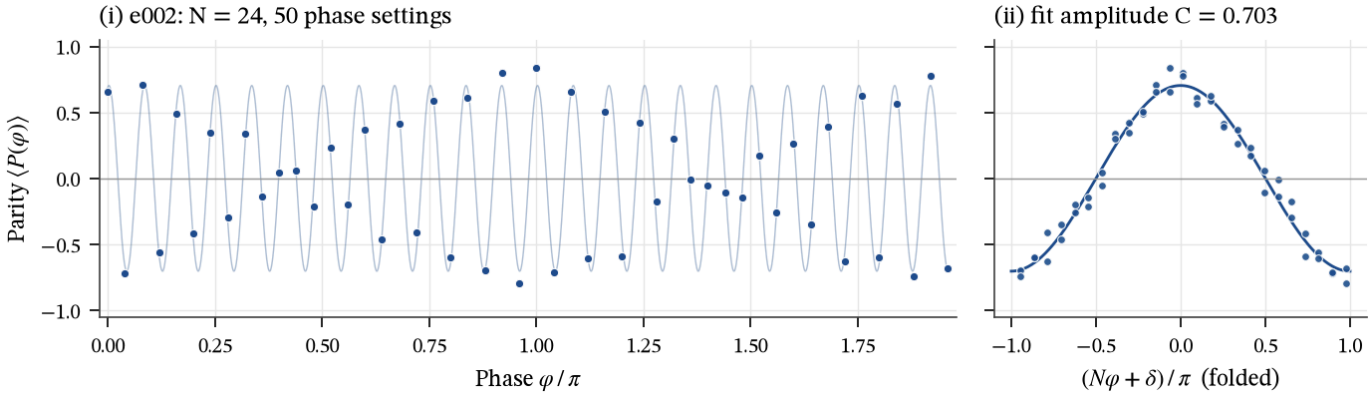


Figure 2. Parity oscillation of e002 ($N = 24$, hardware). (i) Measured parity at each phase φ_j with the fitted $C\cos(N\varphi + \delta)$. (ii) The same points folded onto one period.

Table 1. All experiments in chronological order. HW = ibm_fez; sim = noisy simulator; twin = digital-twin prediction recorded before submission. Layout "(−k)" marks k qubits excluded by the agent. P, C, F as in Eqs. (1)–(3); CI is the bootstrap confidence interval; ✓ = certified ($F_{\text{low}} > 1/2$). Shaded rows are the records.

ID	Mode	N	Layout	2q depth	P	C	F	CI	GME	QPU s
e001	HW	16	weighted_tree	6	0.829	0.852	**0.841**	[0.820, 0.862]	✓	13.00
e001-twin	twin	16	weighted_tree	6	–	–	**0.906**	[0.890, 0.921]	✓	–
e002	HW	24	weighted_tree	8	0.743	0.703	**0.723**	[0.700, 0.746]	✓	18.00
e002-twin	twin	24	weighted_tree	8	–	–	**0.866**	[0.843, 0.885]	✓	–
e003	HW	24	best_chain / linear	23	0.136	0.079	**0.107**	[0.031, 0.163]	✗	18.00
e003-twin	twin	24	best_chain / linear	23	–	–	**0.705**	[0.685, 0.727]	✓	–
e004	HW	32	weighted_tree	9	0.574	0.500	**0.537**	[0.400, 0.641]	✗	22.00
e004-twin	twin	32	weighted_tree	9	–	–	**0.796**	[0.768, 0.820]	✓	–
e005	HW	40	weighted_tree	10	0.723	0.585	**0.654**	[0.520, 0.788]	✓	28.00
e005-twin	twin	40	weighted_tree	10	–	–	**0.727**	[0.698, 0.753]	✓	–
e006	HW	48	weighted_tree	11	0.487	0.423	**0.455**	[0.344, 0.555]	✗	32.00
e006-twin	twin	48	weighted_tree	11	–	–	**0.648**	[0.614, 0.677]	✓	–
e007	HW	56	weighted_tree	11	0.389	0.249	**0.319**	[0.196, 0.465]	✗	37.00
e007-twin	twin	56	weighted_tree	11	–	–	**0.604**	[0.569, 0.638]	✓	–
e008	HW	60	weighted_tree	11	0.294	0.195	**0.244**	[0.031, 0.394]	✗	39.00
e008-twin	twin	60	weighted_tree	11	–	–	**0.642**	[0.600, 0.682]	✓	–
e009	HW	40	weighted_tree	10	0.367	0.462	**0.414**	[0.326, 0.486]	✗	52.00
e009-twin	twin	40	weighted_tree	10	–	–	**0.726**	[0.705, 0.745]	✓	–
e010	HW	36	weighted_tree	9	0.546	0.432	**0.489**	[0.327, 0.609]	✗	25.00
e010-twin	twin	36	weighted_tree	9	–	–	**0.776**	[0.748, 0.803]	✓	–
e011	HW	24	bfs_tree	7	0.893	0.790	**0.842**	[0.720, 0.972]	✓	18.00
e011-twin	twin	24	bfs_tree	7	–	–	**0.845**	[0.822, 0.867]	✓	–
e012	HW	28	weighted_tree	9	0.593	0.566	**0.580**	[0.444, 0.703]	✗	20.00
e012-twin	twin	28	weighted_tree	9	–	–	**0.815**	[0.792, 0.837]	✓	–

5 DISCUSSION AND LIMITATIONS

The statistical confidence interval for the certified fidelity reflects shot noise alone and does not account for potential drift between the 51 circuits of the experiment or between distinct experimental runs. The lower bound of the certified fidelity interval exceeds 1/2 by 0.200. A further consideration is the selection bias introduced by the autonomous agent, which chose configurations and designated a record from multiple noisy estimates. Four hardware experiments were conducted within four qubits of the record size, and reporting the best result from several noisy estimates inherently biases the reported fidelity upward. No correction for this selection effect was applied. The potential for such bias is not hypothetical; an apparent N = 40 record in this run failed its pre-registered confirmation and was subsequently withdrawn. The N = 24 record itself was certified in 2 separate experiments using different tree layouts, sharing 20 of 24 qubits. This mitigates, but does not eliminate, the concern regarding selection bias, as neither of these certification experiments was re-run.

Readout-error mitigation was applied in 100% of experiments. This technique reweights measured populations and parities using the inverse of a per-qubit confusion model. Consequently, the certified fidelity represents a property of the mitigated estimator rather than of raw measurement counts. The model assumes readout errors are uncorrelated across qubits and free of crosstalk. Furthermore, the error rates are estimated from only two calibration circuits, and their associated shot noise is not propagated into the bootstrap interval, which suggests the interval may be somewhat too narrow. A conservative interpretation of the result would necessitate the unmitigated fidelity also exceeding 1/2, a condition not separately established.

The digital twin consistently overestimated hardware fidelity across all matched sizes, with discrepancies ranging from 0.02 to 0.40. The noise model employed for the twin is derived from a single calibration snapshot and omits factors such as crosstalk, leakage, non-Markovian dephasing, and temporal drift. Therefore, the observed gap between twin predictions and hardware performance is expected to increase with N and with circuit duration. Since the twin predictions were finalized prior to the execution of hardware

jobs, the magnitude of this discrepancy provides a direct and unbiased measure of the portion of the device's error budget not captured by the model.

The experimental run spanned 6.7 hours, during which device parameters may have drifted from the initial calibration snapshot. While a fidelity exceeding 1/2 certifies genuine multipartite entanglement, it does not provide a full characterization of the quantum state. Furthermore, the validity of the witness depends on the assumption that the measurements performed correspond to the intended ones. Finally, the agent's search strategy was characterized by its brevity and greedy nature, exploring only a limited portion of the parameter space for roots, layouts, and settings. This suggests that the certification of larger entangled states on this device remains plausible.

6 REPRODUCIBILITY

Run ID	`20261107-220808-qpu`
Run mode / backend	qpu / ibm_fez
Run start / end	2026-11-08 03:08 UTC / 2026-11-08 09:52 UTC
Calibration	live backend query, 2026-11-08T03:07:49+00:00
Budget used	322.0 / 480 QPU s (self-imposed cap; open plan 600 s); LLM calls capped and metered
Agent model	gemini:gemini-2.5-flash
Software	qiskit 2.6.2, qiskit-ibm-runtime 0.51.0, qiskit-aer 0.18.2, numpy 2.5.4

Hardware job IDs:

e001: `dbnuf68ou4as73b57070`; e002: `dbnuf6oqtr0c73admn4g`; e003: `dbnuf7dt7fqc73ak8im0`; e004: `dbnuf7oqtr0c73admn7g`; e005: `dbnuf8dt7fqc73ak8ing`; e006: `dbnuf8tt7fqc73ak8ip0`; e007: `dbnuf9gou4as73b570fg`; e008: `dbnufa8qtr0c73admne0`; e009: `dbo4738ou4as73b5fts0`; e010: `dbo473tt7fqc73akhc70`; e011: `dbo47444148s7391u70g`; e012: `dbo474k4148s7391u71g`

Every number in this paper is computed from `runs/20261107-220808-qpu/` (`calibration.json`, `events.jsonl`, `experiments/*.json`) and can be regenerated with `uv run python -m scripts.write_paper runs/20261107-220808-qpu`.

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